Analysis of a Three-Phase Fully Controlled Full-Wave Rectifier

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Abstract

This paper analyzes the operation of a three-phase, fully controlled, six-pulse rectifier supplying resistive and inductive loads. The firing sequence and phase-voltage intersections are used to describe thyristor commutation and the resulting line-to-line output voltage. Analytical expressions for the average DC output voltage are derived for firing angles below and above 60 degrees, including the effects of a freewheeling diode and inductive-current extinction. Device blocking voltage and current stresses are then discussed, and candidate thyristor and diode specifications reported in the source material are summarized. The derivations identify the assumptions and missing load data that limit numerical current-rating verification.

1 Introduction

This paper examines a three-phase, fully controlled, full-wave rectifier supplied by a balanced three-phase source. The circuit topology is shown in Fig. 1, and the phase voltages applied to the rectifier are shown in Fig. 2. The phase voltages are separated by 120 degrees. The source specification corresponds to a phase-voltage peak of

$$V_{\mathrm{ph,pk}} = \frac{220\sqrt{2}}{\sqrt{3}}\ \mathrm{V}, \tag{1}$$

which gives a line-to-line peak voltage of approximately 311.12 V.

Unlike an uncontrolled or semi-controlled bridge, this converter uses six thyristors, allowing the average DC output voltage to be adjusted by delaying their gate signals. Until a suitable thyristor pair is triggered, the load is not connected to the source through the bridge. Under continuous-current operation, one upper-group and one lower-group thyristor conduct at a time. The firing sequence is T6–T1, T1–T2, T2–T3, T3–T4, T4–T5, and T5–T6.

Thyristors T1, T3, and T5 are connected to phases R, S, and T, respectively. Each conducts for 120 degrees in the continuous-current operating mode. The two devices in the same phase leg are separated by 180 degrees. Successive commutations occur every 60 degrees. Unless otherwise stated, the switching discussion below uses a firing delay of $\alpha = 30^\circ$.

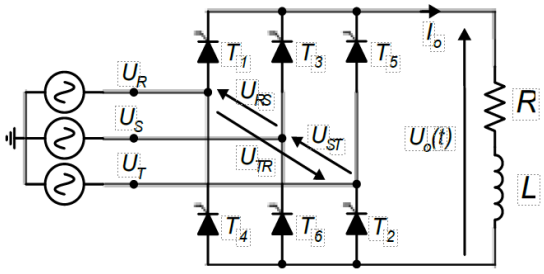


Figure 1: Circuit diagram of the fully controlled three-phase full-wave rectifier with an inductive load.

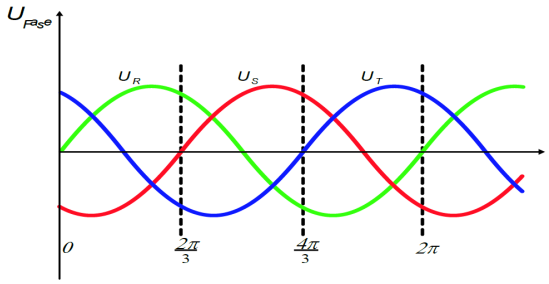


Figure 2: Three-phase source voltages supplying the rectifier.

2 Rectifier Operation

Table 1 lists the natural commutation angles at which the phase-voltage ordering changes. The firing delay α is measured from the corresponding base angle. For $\alpha = 30°$, T6 is triggered at 0 degrees and T1 at 60 degrees. Before T1 is triggered, T6 alone can conduct; after T1 is triggered, the load is connected across U_{RS}, as shown in Fig. 3.

At 90 degrees, the phase ordering changes and T2 becomes forward-biased. It does not conduct until its gate pulse arrives 30 degrees later. During this delay, the inductive load current maintains the existing current path. At 120 degrees, T2 turns on and the conducting pair changes to T1–T2; the load then receives U_{RT} (Fig. 4).

At 150 degrees, T3 becomes forward-biased, but T1 continues conducting until T3 is triggered at

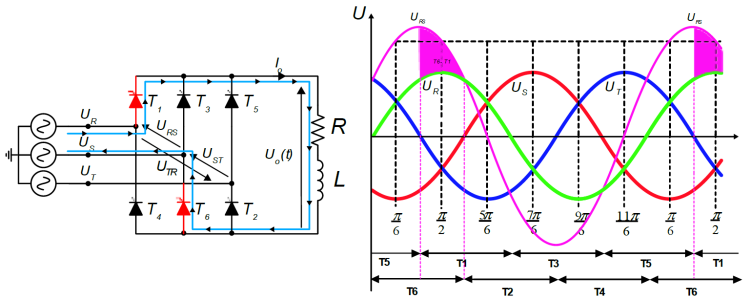


Figure 3: Load voltage during conduction of thyristors T1 and T6.

Table 1: Natural commutation angles of the thyristors

Thyristor	Base angle (°)	Base angle (rad)
T1	30°	$\frac{\pi}{6}$
T2	90°	$\frac{\pi}{2}$
T3	150°	$\frac{5\pi}{6}$
T4	210°	$\frac{7\pi}{6}$
T5	270°	$\frac{3\pi}{2}$
T6	330°	$\frac{11\pi}{6}$

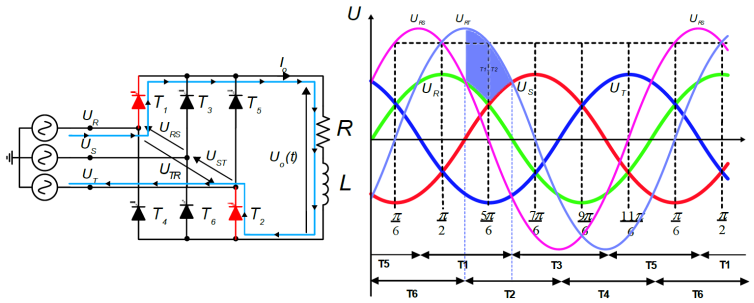


Figure 4: Load voltage during conduction of thyristors T1 and T2.

180 degrees. The conducting pair then becomes T3–T2 and the load voltage is U_{ST} (Fig. 5). At 210 degrees, T4 becomes forward biased; when it is triggered at 240 degrees, the pair T3–T4 applies U_{SR} to the load (Fig. 6).

At 270 degrees, T5 becomes forward-biased. Its gate pulse arrives at 300 degrees, changing the conducting pair to T5–T4 and the load voltage to U_{TR} (Fig. 7). At 330 degrees, T6 becomes forward biased; it is triggered at 360 degrees, and the pair T5–T6 applies U_{TS} to the load (Fig. 8). The sequence then repeats. The resulting output-voltage waveform is shown in Fig. 9.

For a purely resistive load, the load current follows the output-voltage waveform and is given by $i_o(t) = v_o(t)/R_L$. For an inductive load, the current depends on both resistance and inductance; it need not have the same shape as the voltage. The numerical load resistance is not specified in the source document, so a numerical output-current waveform cannot be reconstructed from the supplied information. The simulated current waveform is referenced in Fig. 10.

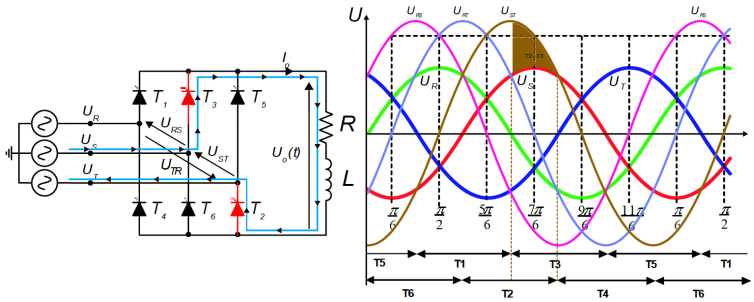


Figure 5: Load voltage during conduction of thyristors T3 and T2.

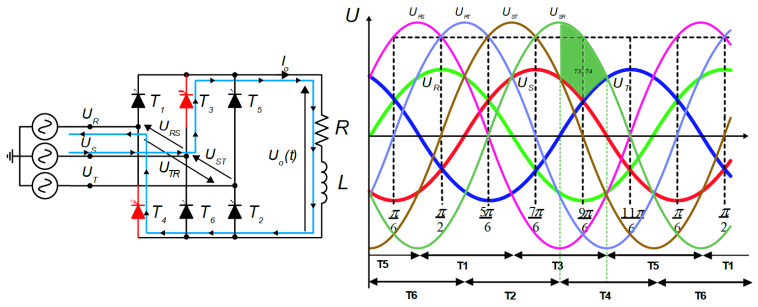


Figure 6: Load voltage during conduction of thyristors T3 and T4.

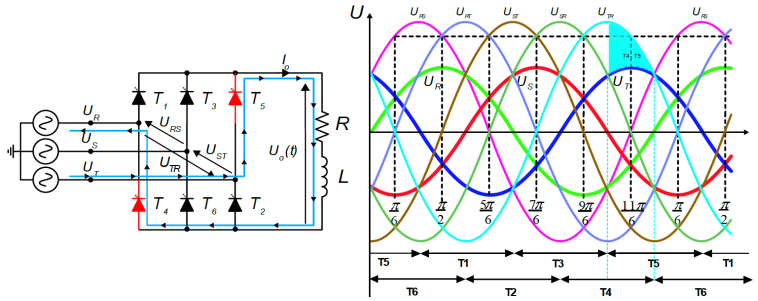


Figure 7: Load voltage during conduction of thyristors T5 and T4.

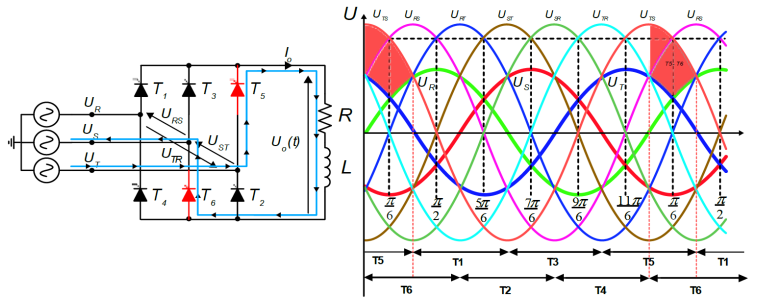


Figure 8: Load voltage during conduction of thyristors T5 and T6.

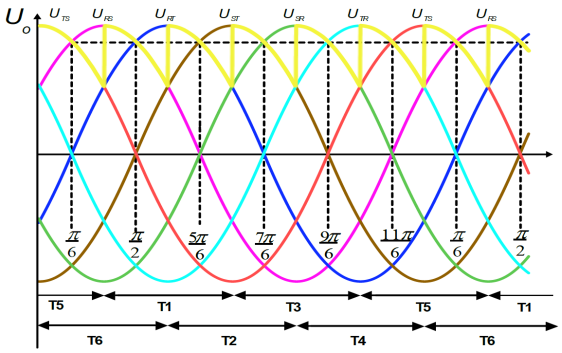


Figure 9: Rectifier load voltage over one operating cycle.

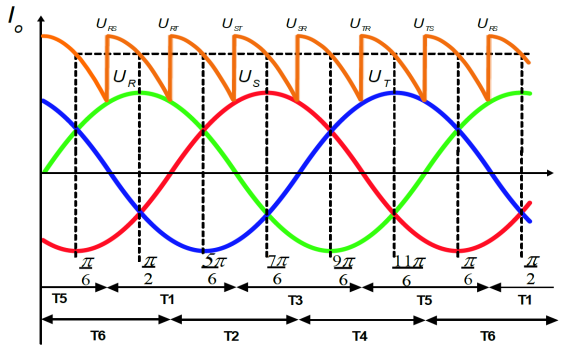


Figure 10: Load current over one operating cycle.

3 Mathematical Analysis

3.1 Phase-Voltage Intersections

The natural commutation angles can be calculated by equating the phase voltages. Let $\theta = \omega t$, and define

$$v_R(\theta) = V_m \sin\theta, \tag{2}$$

$$v_S(\theta) = V_m \sin\left(\theta - \frac{2\pi}{3}\right), \tag{3}$$

$$v_T(\theta) = V_m \sin\left(\theta - \frac{4\pi}{3}\right). \tag{4}$$

For phases R and S, setting $v_R = v_S$ gives

$$\sin\theta = \sin\left(\theta - \frac{2\pi}{3}\right), \tag{5}$$

$$\theta = \frac{5\pi}{6} + k\pi, \tag{6}$$

so their intersections occur at 150 and 330 degrees modulo 360 degrees. For phases S and T,

$$\sin\left(\theta - \frac{2\pi}{3}\right) = \sin\left(\theta - \frac{4\pi}{3}\right), \tag{7}$$

$$\theta = \frac{\pi}{2} + k\pi, \tag{8}$$

which gives 90 and 270 degrees. For phases T and R,

$$\sin\left(\theta - \frac{4\pi}{3}\right) = \sin\theta, \tag{9}$$

$$\theta = \frac{\pi}{6} + k\pi, \tag{10}$$

which gives 30 and 210 degrees. These results agree with the commutation angles in Table 1. Angles in the following integrals are expressed in radians.

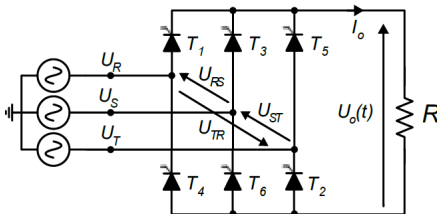


Figure 11: Circuit diagram of the fully controlled three-phase rectifier with a resistive load.

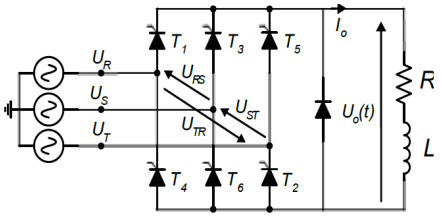


Figure 12: Circuit diagram of the fully controlled three-phase rectifier with an inductive load and a freewheeling diode.

3.2 Average Output Voltage

3.2.1 Load Configurations

The resistive-load circuit is referenced in Fig. 11; the inductive-load circuit is shown in Fig. 1; and the inductive-load circuit with a freewheeling diode is referenced in Fig. 12.

Unlike a diode bridge, a thyristor does not turn off as soon as it becomes reverse-biased. It continues to conduct until its current falls below the holding current. With a resistive load, current falls to zero when the applied voltage crosses zero. With an inductive load, stored magnetic energy can sustain current beyond the voltage zero crossing, extending thyristor conduction and introducing a negative-voltage interval. A freewheeling diode provides an alternate path for inductive current and prevents the load voltage from becoming negative during that interval.

The line voltage U_{RS} crosses zero 120 degrees after its natural base angle. With a firing delay below 60 degrees, the next commutation occurs before this zero crossing. At 60 degrees, the firing instant coincides with the zero crossing; this is the boundary between the two operating cases analyzed below.

3.2.2 Firing Angles Below 60 Degrees

For $0 \leq \alpha < 60°$, the load-voltage waveform has a 60-degree period, $T = \pi/3$, and the same average-voltage expression applies to the resistive and inductive cases considered here. The output segment associated with U_{RS} starts at $\pi/6 + \alpha$ and ends at $\pi/2 + \alpha$. Let $V_{LL,\mathrm{pk}}$ denote the peak line-to-line

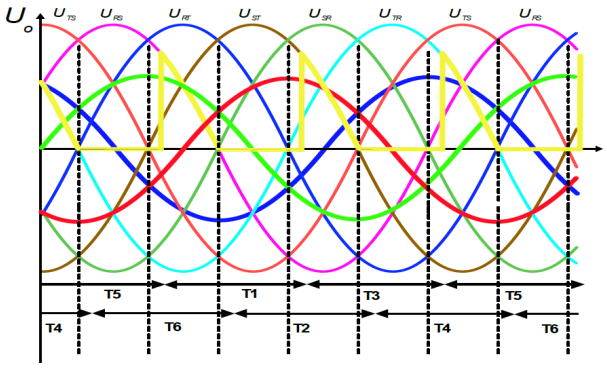


Figure 13: Rectifier load voltage for a firing angle greater than 60 degrees.

voltage, so that $u_{RS}(\theta) = V_{LL,\text{pk}} \sin(\theta + \pi/6)$. The average output voltage is

$$V_{o,\text{avg}} = \frac{1}{T} \int_0^T v_o(\theta)\, d\theta \tag{11}$$

$$= \frac{3V_{LL,\text{pk}}}{\pi} \int_{\pi/6+\alpha}^{\pi/2+\alpha} \sin\left(\theta + \frac{\pi}{6}\right)\, d\theta \tag{12}$$

$$= \frac{3V_{LL,\text{pk}}}{\pi} \cos\alpha \tag{13}$$

$$= \frac{3\sqrt{2}V_{LL,\text{rms}}}{\pi} \cos\alpha. \tag{14}$$

3.2.3 Firing Angles Above 60 Degrees

For $\alpha > 60°$, a resistive load and an inductive load with a freewheeling diode have zero-voltage intervals. The output waveform for this case is referenced in Fig. 13. For each 60-degree output period, conduction begins at $\pi/6 + \alpha$ and ends when U_{RS} crosses zero at $5\pi/6$. The average is obtained by integrating the nonzero segment over a period $T = \pi/3$:

$$V_{o,\text{avg}} = \frac{3V_{LL,\text{pk}}}{\pi} \int_{\pi/6+\alpha}^{5\pi/6} \sin\left(\theta + \frac{\pi}{6}\right)\, d\theta \tag{15}$$

$$= \frac{3V_{LL,\text{pk}}}{\pi} \left[1 + \cos\left(\alpha + \frac{\pi}{3}\right)\right]. \tag{16}$$

For an inductive load without a freewheeling diode, the current may persist beyond the line-voltage zero crossing. Let θ_e denote the additional extinction angle determined by the load and its current. The waveform is referenced in Fig. 14. Evaluating the line-voltage segment used in the source derivation and averaging over the 60-degree output period gives

$$V_{o,\text{avg}} = \frac{3V_{LL,\text{pk}}}{\pi} \int_{\pi/6+\alpha}^{5\pi/6+\theta_e} \sin\left(\theta + \frac{\pi}{6}\right)\, d\theta \tag{17}$$

$$= \frac{3V_{LL,\text{pk}}}{\pi} \left[\cos\theta_e + \cos\left(\alpha + \frac{\pi}{3}\right)\right]. \tag{18}$$

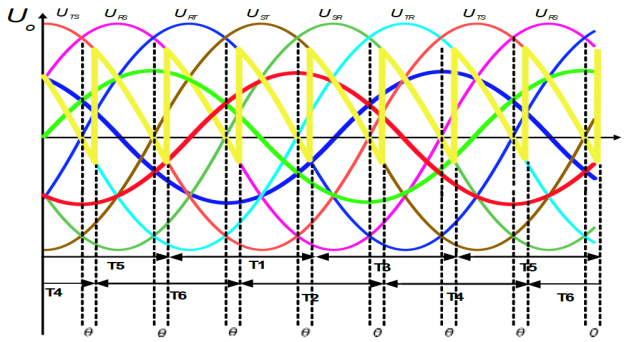


Figure 14: Inductive-load voltage waveform for a firing angle greater than 60 degrees.

4 Semiconductor Stress

4.1 Thyristor Conduction Sequence

The following intervals summarize the simulated operation for $\alpha = 30°$. The applied load voltage and the off-state voltage across T1 are listed for each 60-degree interval. The corresponding waveforms are referenced in Fig. 15.

Table 2: Output voltage and T1 stress over one cycle

Interval	Load voltage	T1 state or blocking voltage
0°–60°	U_{TS}	Off; blocks U_{RT}
60°–120°	U_{RS}	Conducting
120°–180°	U_{RT}	Conducting
180°–240°	U_{ST}	Off; blocks U_{RS}
240°–300°	U_{SR}	Off; blocks U_{RS}
300°–360°	U_{TR}	Off; blocks U_{RT}

In the first 60-degree interval, T5 and T6 establish the load-current path while T1 blocks the difference between phases R and T. When T1 is triggered at 60 degrees, it conducts, and its terminal voltage is approximately zero in the ideal-switch model. It remains on through the subsequent commutation of the lower-group device and turns off when T3 is triggered at 180 degrees. It then blocks the phase-to-phase voltage associated with phase R. The same pattern repeats with the corresponding phase shifts for the other thyristors. Lower-group devices are shifted by 180 degrees relative to the upper-group devices.

4.2 Voltage and Current Ratings

T1 is connected to phase R and can block line-to-line voltages involving that phase, including U_{RS} and U_{RT}. The source gives a maximum line-to-line voltage of

$$V_{LL,\mathrm{pk}} = 220\sqrt{2}\ \mathrm{V} \approx 311.12\ \mathrm{V}. \tag{19}$$

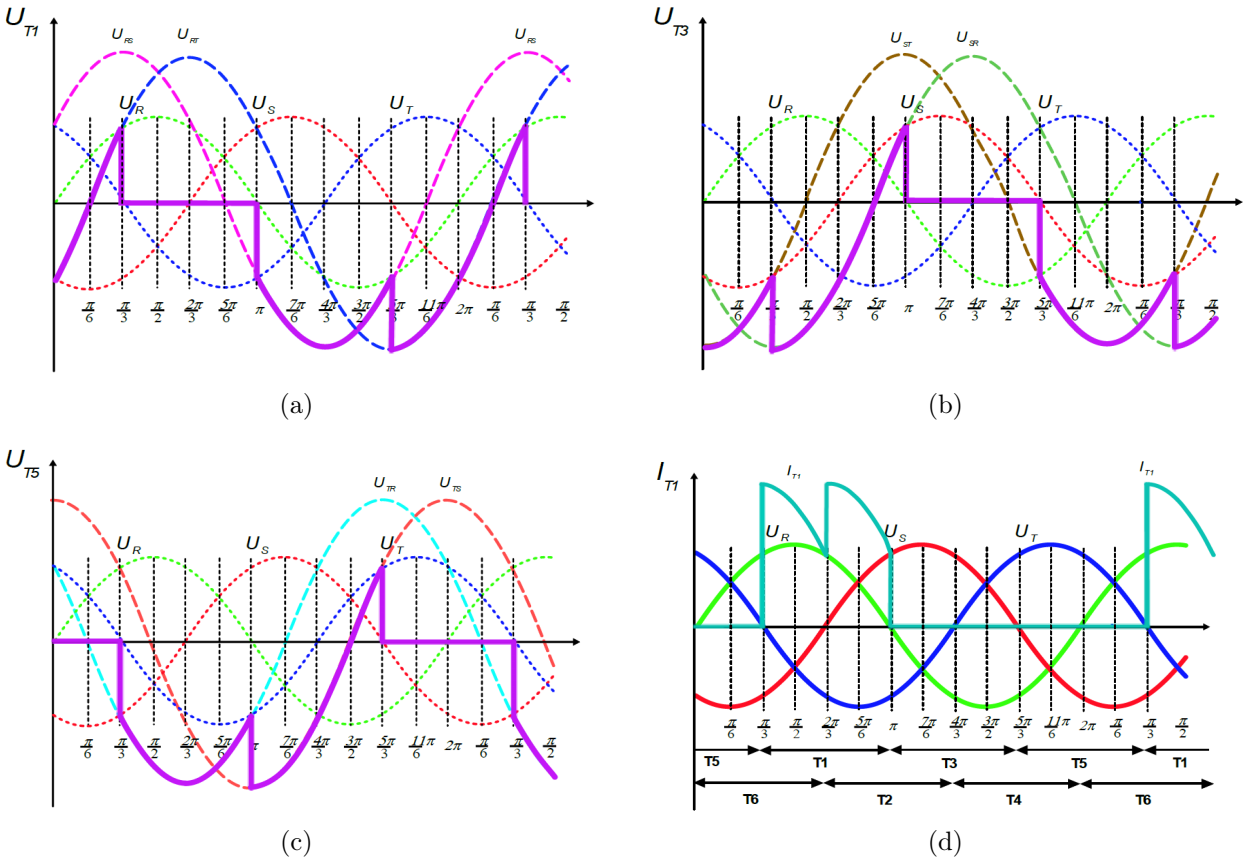


Figure 15: Thyristor waveforms. Blocking-voltage waveform across thyristor: a) T1, b) T3, c) T5 and d) Current through T1

The same blocking-voltage magnitude applies to the other thyristors, with phase shifts determined by their connections. The source proposes a minimum nominal blocking rating of approximately 320 V; practical selection must also account for transients, tolerances, thermal conditions, and the manufacturer's specified repetitive ratings.

When a thyristor is on, its idealized blocking voltage is zero and load current flows through it. When it is off, its current is approximately zero while it blocks the relevant line-to-line voltage. The thyristor current is set by the load and cannot be inferred by equating its numerical value to the voltage. In particular, 311.12 is a voltage in volts, not a current in amperes. Because the source does not provide the load resistance or sufficient inductive-load parameters, the required thyristor current rating cannot be verified from the available data.

The gate-triggering sequence over one rectification cycle is summarized in Fig. 16.

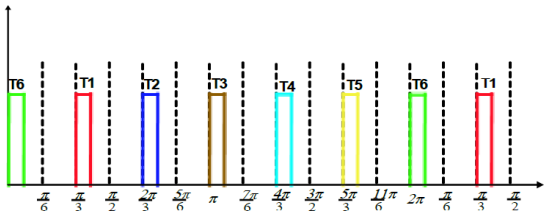


Figure 16: Thyristor firing sequence during one rectification cycle.

4.3 Freewheeling-Diode Stress

For $\alpha = 30°$, the source analysis predicts that the freewheeling diode is not substantially forward biased under the stated operating conditions. It therefore blocks voltage while its current remains approximately zero. When $\alpha > 60°$, inductive current can circulate through the diode after the source-to-load path is interrupted. The diode then clamps the load voltage, and its reverse-voltage requirement is comparable in magnitude to the peak line-to-line voltage. The simulated diode-voltage waveforms are referenced in Figs. 17 and 18.

The diode-current discussion in the source considers two different inductance values. Fig. 19 is described for 0.1 μH, whereas Fig. 20 is described for 0.1 mH and reports a simulated peak current of approximately 12 A. These values are not interchangeable; the actual diode-current rating must

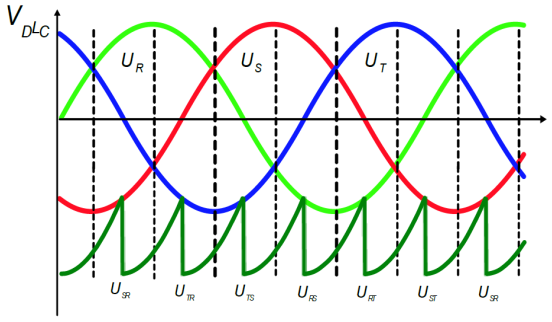


Figure 17: Blocking-voltage waveform across the freewheeling diode.

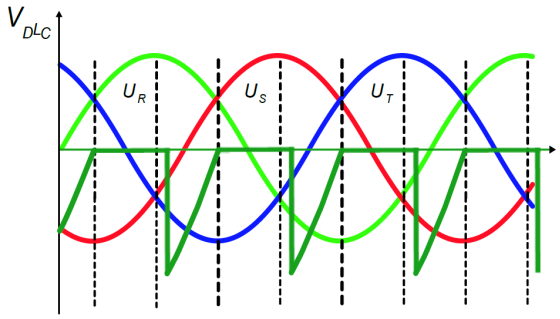


Figure 18: Freewheeling-diode blocking voltage for a firing angle greater than 60 degrees.

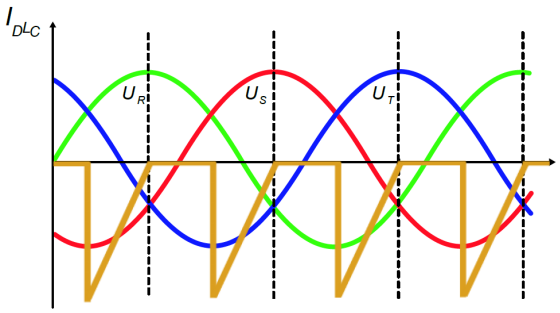


Figure 19: Freewheeling-diode current for a firing angle greater than 60 degrees and an inductance of 0.1 μH.

be established using the inductance and operating conditions of the intended circuit. The source recommends approximately 320 V and 15 A as minimum selection targets for its example, subject to verification against the actual transient and thermal requirements.

5 Commercial Semiconductor Candidates

The source lists the following commercial components and ratings. They are reproduced as reported and should be checked against the manufacturer's current datasheets before design use.

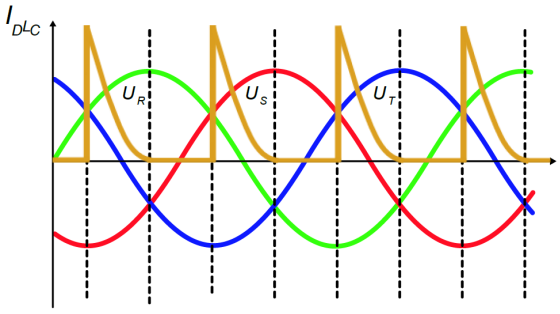


Figure 20: Freewheeling-diode current for a firing angle greater than 60 degrees and an inductance of 0.1 mH.

5.1 Q4035NH5RP Device

The listed Littelfuse Q4035NH5RP is identified in the source as a triac, not as a conventional unidirectional SCR. Its suitability for a six-thyristor bridge must therefore be verified before selection.

Table 3: Q4035NH5RP specifications reported in the source

Parameter	Reported value
Manufacturer and series	Littelfuse; Qxx35xHx
Device category	Triac
RMS on-state current	35 A
Non-repetitive surge current	290 A and 350 A (conditions not stated)
Repetitive off-state voltage, V_{DRM}	400 V
Off-state leakage current, I_{DRM}	10 μA at V_{DRM}
On-state voltage	1.5 V
Maximum holding current, I_H	75 mA
Gate trigger voltage, V_{GT}	2 V
Gate trigger current, I_{GT}	50 mA
Operating temperature	-40°C to $+125$°C
Package and mounting	SMTO-263-2; surface mount

5.2 IDH12SG60C Diode

The listed Infineon IDH12SG60C is a silicon-carbide Schottky diode in a through-hole TO-220-2 package. The reported specifications are summarized in Table 4.

Table 4: IDH12SG60C specifications reported in the source

Parameter	Reported value
Manufacturer and series	Infineon Technologies; IDH12SG60
Technology	Silicon carbide (SiC)
Average forward current, I_F	12 A
Repetitive reverse voltage, V_{RRM}	600 V
Forward voltage, V_F	1.8 V
Non-repetitive surge current, I_{FSM}	59 A
Reverse current, I_R	1 μA
Operating temperature	-55°C to $+175$°C
Package and mounting	TO-220-2; through-hole
Maximum power dissipation	125 W
Maximum dimensions	9.45 mm high; 10.2 mm long